

5.06 Chi-squared Tests

5.06a	Contingency tables	a) Be able to use a chi-squared (χ^2) test with the appropriate number of degrees of freedom to test for independence in a contingency table and interpret the results of such a test. <i>Rows or columns, as appropriate, should be combined so that each expected frequency is at least 5, and Yates' correction should be used in the special case of a 2×2 table.</i> <i>A table of critical values of the χ^2 distribution will be provided.</i> <i>Includes calculation of expected frequencies and contributions to the test statistic.</i> <i>Questions may require candidates to calculate some expected frequencies and contributions to the test statistic, but will not involve lengthy calculations.</i>	
5.06b 5.06c	Fitting a theoretical distribution	b) Be able to fit a theoretical distribution, as prescribed by a given hypothesis involving a given ratio, proportion or discrete uniform distribution, to given data. <i>Questions may require candidates to calculate some expected frequencies, but will not involve lengthy calculations.</i>	c) Extend their knowledge of fitting distributions to other known or given discrete and continuous distributions. <i>Questions may require candidates to calculate some expected frequencies, but will not involve lengthy calculations.</i>
5.06d	Goodness of fit test	d) Be able to use a χ^2 test with the appropriate number of degrees of freedom to carry out the corresponding goodness of fit test. <i>Where necessary, adjacent classes should be combined so that each expected frequency is at least 5.</i> <i>A table of critical values of the χ^2 distribution will be provided.</i>	